

Harshith Kumar Kakkirala

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[GitHub](#) | [LinkedIn](#) | [Portfolio](#)

About Me

A Computer Science undergraduate majoring in Data Science, equipped with core skills in machine learning, full-stack engineering, and software architecture. Skilled at developing AI-driven applications that scale across both web and mobile environments. Driven by a desire to solve practical problems through intelligent, efficient systems, and currently pursuing a Software Engineer position to contribute toward impactful technology solutions.

Skills

Programming	Python, Java, C/C++, JavaScript
Frameworks & Libraries	TensorFlow, Scikit-learn, LangChain
Full Stack	HTML, CSS, MERN Stack (MongoDB, Express.js, React, Node.js), REST APIs, Microservices
Databases	MySQL, PostgreSQL, Firebase
Tools & Platforms	Git, GitHub, Docker, Vercel, Figma
Soft Skills	Analytical Thinking, Problem Solving, Leadership, Team Collaboration

Projects

AgriChain

Sept 2025

- Tech Stack:** Next.js, React, TypeScript, Firebase Realtime Database, Ethereum/Polygon, Solidity, ethers.js
- Engineered a blockchain-enabled agricultural supply chain traceability platform with role-based access for Farmers, Aggregators, Retailers, and Consumers.
 - Implemented QR-based batch creation and real-time camera scanning to log trace events (transport, processing, storage, retail) across the supply chain.
 - Connected Solidity smart contracts on the Polygon testnet via ethers.js to provide optional tamper-proof blockchain verification of product history.

GenSathi

Feb 2026

- Tech Stack:** Next.js, React, Firebase Firestore, Tailwind CSS, Leaflet
- Developed a civic engagement platform connecting citizens and city administrators for real-time issue reporting, tracking, and resolution.
 - Built a smart duplicate-detection system using the Haversine distance formula to flag similar complaints within a 150m radius and redirect users to upvote instead.
 - Designed an XP-based gamification and community upvoting engine to prioritize critical issues, paired with proof-based (before/after photo) resolution workflows for admins.

CampusPocket

May 2026

- Tech Stack:** React Native, Expo, Supabase, PostgreSQL, Groq API (Llama 3 8B), NativeWind
- Built a full-stack mobile EdTech platform connecting schools, parents, and students with real-time academic insights, automated scheduling, and performance analytics.
 - Engineered a custom Linear Regression Growth Modeling (LRGM) algorithm to analyze multi-year academic data and predict student performance trends.
 - Integrated an AI-powered curriculum chatbot and parent performance summaries via Groq API, using minimal RAG and 24-hour response caching to cut token usage and API costs.

Education

VNR Vignana Jyothi Institute of Engineering & Technology, Hyderabad, Telangana **2024–2028**
B.Tech in Computer Science and Engineering (Honors), Specialization in Data Science CGPA: 9.1
Also pursuing **B.S. in Data Science and Applications (Diploma Level)**, IIT Madras CGPA: 8.2
Viswasanthi E.M High School, Vuyyuru, Andhra Pradesh

Intermediate (CBSE) – PCM
Matriculation (CBSE)

91.8% | July 2024
95.6% | May 2022

Publications

- Kakkirala Harshith Kumar et al., “[Optimizing Water Quality Monitoring Using Principal Component Analysis: A Case Study of Hyderabad’s Water System](#),” *Recent Patents on Engineering* (Bentham Science Publishers) – Q3, Scopus Indexed, 2025.
- Kakkirala Harshith Kumar et al., “Blockchain for Transparent and Traceable Agri-Food Supply Chains,” in *Blockchain Beyond Cryptocurrency: Innovations and Applications*, Chapter 7, pp. 1–44 – **In Progress**, 2026.

Achievements

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|---|-----------------|
| Runner – VJ Hackathon 2025
Organized by CSI, VNRVJIET | Oct 2025 |
| 1st Place – Project Contest (Software), CONVERGENCE 2K25
National Technical Symposium, VNRVJIET | Nov 2025 |
| 2nd Place – SynthVision Hackathon
Part of AI Week, organized by Krithomedh, VNRVJIET | Feb 2026 |

Leadership & Activities

Computer Society of India (CSI), VNRVJIET

- Organized **Navigator**, a career guidance orientation session for **300+ junior students**, covering technical domains, internships, and academic opportunities.
- Served as **Point of Contact (POC)** for **RE:VERSE – “Turn the Page, Change the Fate”**, an immersive mystery-puzzle event for first-years, coordinating execution for **100+ participating teams**.

VJ DataQuesters – Data Science and Analytics Club

- Design volunteer, actively creating posters and promotional graphics for club festivals and events.

Declaration

I hereby declare that the details and information provided above are complete and true to the best of my knowledge. I understand that withholding or misstating anything may result in refusal for placement.